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0.1 Experimental Evidence Overview

This section describes the experimental evidence that supports the analyses presented in this chapter. The evidence is organised into three categories: classification experiments, segmentation experiments, and mobile-deployment benchmarks. Each category is further subdivided into primary retained results and supplementary supporting evidence.

0.1.1 Classification Evidence

The primary classification evidence consists of machine-readable result files, configuration logs, and trained checkpoints produced under a fixed seed 42. The SDI-4 dataset, comprising 1,149 images across four classes (Healthy, BG, WSSV, WSSV_BG), was partitioned into training, validation, and test sets at an approximate 70:15:15 ratio (804/172/173). The final proposed classifier—YOLO26m-cls augmented with ASL-LDAM and SimAM-DCFR—achieved a Macro-F1 of 91.01%, accuracy of 91.33%, and Cohen’s Kappa of 0.8816 on the SDI-4 test partition. The cross-entropy (CE) baseline, which shares the same YOLO26m-cls backbone and training protocol without the custom attention and loss modifications, attained 89.02% Macro-F1 and 89.02% accuracy.

Supplementary classification evidence includes:

- **YOLO baseline screening:** A controlled comparison of 15 YOLO-cls variants (YOLOv8, YOLO11, YOLO26) at the nano, small, medium, large, and extra-large scales under the CE baseline protocol. The retained machine-readable CSV (yolo_baseline_comp

arison_seed42.csv) reports Macro-F1, accuracy, Cohen’s Kappa, parameter count, and model size for each variant.

- **TIMM baseline screening:** A separate benchmark of 17 classification architectures from the TIMM library (including ConvNeXt, MobileNet, EfficientNet, and lightweight variants) under the same SDI-4 partition and CE protocol. Results are recorded in `timm_baseline_comparison_seed42.csv`.
- **Component analysis:** Attention-benchmark logs from an RTX 4090 comparing the ASL-LDAM, SimAM-DCFR, and combined configurations against the CE baseline. These are available under the archived evidence: `CVio_SDI4_ASL_LDAM_Attention_Benchmark_RTX4090.zip` and the CE counterpart.
- **Additional-source evaluation:** The EXT-3-Original dataset (315 images, three classes) and the integrated SDI-4 + EXT-3-Original experiment, with partition-wise merged training, validation, and test splits. Retained results include class-level confusion matrices and source-stratified accuracy.
- **Kaggle T4 inference:** GPU-accelerated inference benchmarks recorded in `CVio_Final_Proposed_T4_Seed42_FINAL.zip` and `CVio_Classification_Kaggle_T4x2_Evidence_2026-08-11.zip`.

0.1.2 Segmentation Evidence

Segmentation experiments were conducted by team member Nguyen Van Phong. The primary retained evidence for the segmentation analysis includes:

- Baseline YOLO11n-seg results on the shrimp disease segmentation dataset, with masks for BG and WSSV classes (Healthy images are included as negative samples with no positive mask polygons).
- Attention-based segmentation variants and Fourier-domain segmentation experiments, evaluated under the same held-out test partition.
- Supplementary robustness and failure-mode analyses documenting known failure cases.

Detailed segmentation results are presented in Sections 0.10 through 0.13. The classification agent does not modify segmentation claims; it reproduces them from the accepted baseline evidence.

0.1.3 Mobile-Deployment Evidence

Mobile deployment evidence is preliminary and comprises TFLite conversion artifacts (FP32 and FP16 variants) and latency measurements obtained under controlled conditions. The retained artifacts include:

- CVio_SD14_Mobile20_seed42_2026-08-10.zip: TFLite delegates and inference logs.
- yolo26m_asl_ldam_simam_dcfr_fp32.tflite: FP32 TFLite model (approximately 39.52 MiB).
- yolo26m_asl_ldam_simam_dcfr_fp16.tflite: FP16 TFLite model (approximately 19.81 MiB).

Mobile evidence is discussed in Section 0.14. The TFLite results are treated as prototype evidence; they do not constitute clinical or real-time deployment guarantees.

0.1.4 Evidence Provenance and Limitations

All primary numerical claims in this chapter are traceable to machine-readable CSV files, configuration logs, or trained checkpoint hashes. The following hash values are recorded for the core SDI-4 checkpoints:

- Proposed SDI-4 checkpoint SHA-256: bf4d969465025e870a61be5e218dfc55b9b2e962dec416f220b1f130703fb1c7
- CE baseline SDI-4 checkpoint SHA-256: 4305e49158129c4a6acaa8fdaf7b5982a7f4d93cc233f11d17479e04b0e438c1

Where evidence is incomplete or where a numerical claim must be inferred from a visual source rather than a machine-readable record, the limitation is stated explicitly. No numerical claim in this chapter is fabricated or inferred from a plot when a machine-readable counterpart should exist.

0.2 Dataset Composition and Evaluation Partitions

This section describes the composition of the datasets used in the classification and segmentation experiments and the corresponding evaluation partitions.

0.2.1 SDI-4 Classification Dataset

The SDI-4 dataset is the primary evaluation benchmark for the classification task. It contains 1,149 RGB images of shrimp, sourced from the ShrimpDiseaseImageBD repository and augmented by the project team. The dataset is organised into four classes:

- **Healthy** (403 images): Shrimp specimens with no visible signs of disease.
- **BG** (198 images): Shrimp specimens exhibiting black gill disease.
- **WSSV** (328 images): Shrimp specimens infected with White Spot Syndrome Virus.

- **WSSV_BG** (220 images): Shrimp specimens exhibiting both WSSV and BG co-infection.

All images were resized to a uniform resolution of 224×224 pixels, consistent with the input requirements of the YOLO-cls architecture.

Partitioning Strategy

The dataset was partitioned using a fixed random seed of 42 to ensure reproducibility. The split was conducted at the image level using a stratified strategy that preserves the class distribution across partitions. The resulting partition sizes are:

$$\text{Train} = 804 \quad (70\%), \quad \text{Validation} = 172 \quad (15\%), \quad \text{Test} = 173 \quad (15\%).$$

The class-level distribution within each partition is reported in Table 1.

Table 1: Class distribution across SDI-4 partitions (seed 42).

Class	Train	Validation	Test	Total
Healthy	282	60	61	403
BG	139	30	29	198
WSSV	229	49	50	328
WSSV_BG	154	33	33	220
Total	804	172	173	1,149

0.2.2 EXT-3-Original Dataset

The EXT-3-Original dataset consists of 315 RGB images sourced from an external collection, covering three classes: Healthy, BG, and WSSV. This dataset does not include the WSSV_BG co-infection class. The images were captured under different environmental conditions and resolutions from the SDI-4 dataset, providing a domain-shift evaluation scenario.

The partition was fixed at:

$$\text{Train} = 221, \quad \text{Validation} = 47, \quad \text{Test} = 47.$$

0.2.3 Integrated SDI-4 + EXT-3-Original Dataset

To evaluate cross-source generalisation, the SDI-4 and EXT-3-Original datasets were merged at the partition level: each dataset was split independently using the same seed 42, and the corresponding partitions (train, validation, test) were concatenated. The resulting integrated dataset has:

$$\text{Train} = 1,025, \quad \text{Validation} = 219, \quad \text{Test} = 220.$$

The integrated test partition comprises images from both sources, with the WSSV_BG class appearing only in the SDI-4 subset. The test composition is reported in Table 2.

Table 2: Integrated test partition composition (SDI-4 + EXT-3-Original).

Class	SDI-4 subset	EXT-3 subset
Healthy	61	11
BG	29	16
WSSV	50	20
WSSV_BG	33	–
Total	173	47

0.2.4 Segmentation Dataset

The segmentation dataset is derived from the same ShrimpDiseaseImageBD source. It contains images with pixel-level polygon annotations for two positive classes: BG and WSSV. Healthy images from the original source are retained in the segmentation training set as negative samples (i.e., images with no positive mask polygons). These images participate in the optimisation process by contributing zero-valued loss for the mask head, but they do not introduce positive mask supervision for the Healthy class.

The segmentation partition sizes and class distribution are documented in the segmentation experimental records (Section 0.10).

0.3 Classification Baseline Screening

The baseline screening stage evaluated candidate classification architectures under a uniform cross-entropy (CE) training protocol on the SDI-4 dataset. The goal was to identify promising backbones for the proposed method while establishing a reproducible reference point for measuring improvement.

0.3.1 YOLO-Cls Baseline

Fifteen YOLO-cls variants from the YOLOv8, YOLO11, and YOLO26 model families were evaluated at scales ranging from nano (approximately 1.5M parameters) to extra-large (approximately 56M parameters). All models were trained using the same CE protocol with identical data partitions (seed 42) and training configuration. The complete results are summarised in Table 3.

The YOLO26m-cls variant was selected as the reference backbone for the proposed method based on the following considerations:

- **Best Macro-F1:** YOLO26m-cls achieved the highest Macro-F1 (89.02%) among all screened variants under the CE protocol.

Table 3: YOLO-cls baseline screening results on SDI-4 test (seed 42).

Model	Params (M)	Macro-F1 (%)	Accuracy (%)	Kappa
YOLO26m-cls	10.35	89.02	89.02	0.8505
YOLO26x-cls	28.34	88.13	89.02	0.8501
YOLOv8m-cls	15.77	87.18	87.86	0.8342
YOLOv8s-cls	5.08	87.10	87.86	0.8343
YOLO11m-cls	10.35	86.75	87.28	0.8263
YOLOv8x-cls	56.13	86.22	86.71	0.8192
YOLOv8l-cls	36.19	85.86	86.71	0.8184
YOLO26l-cls	12.82	85.76	86.13	0.8109
YOLO26n-cls	1.53	84.20	85.55	0.8014
YOLO11x-cls	28.34	83.74	84.97	0.7944
YOLO11s-cls	5.44	83.17	83.82	0.7787
YOLO11n-cls	1.53	82.83	83.82	0.7775
YOLOv8n-cls	1.44	82.71	84.39	0.7858
YOLO26s-cls	5.44	82.15	83.24	0.7713
YOLO11l-cls	12.82	81.23	82.08	0.7560

- **Parameter efficiency:** At 10.35M parameters, YOLO26m-cls offers a favourable trade-off between capacity and computational cost relative to the larger YOLO26x (28.34M) and YOLOv8x (56.13M) variants. The improvement over YOLO26m-cls is marginal, while the parameter count more than doubles.
- **Engineering preference:** The final mobile-oriented target is approximately 15M parameters or fewer. YOLO26m-cls at 10.35M (approximately 19.92 MiB as FP32) leaves room for the additional ASL-LDAM and SimAM-DCFR components while remaining within the lightweight target.

0.3.2 TIMM Baseline

Seventeen additional architectures from the TIMM library were evaluated under the same seed-42 partition and CE protocol. The results are reported in Table 4.

The TIMM baseline confirms that ConvNeXt-Tiny (IN22K) achieves the highest Macro-F1 among screened TIMM architectures (84.16%), but at 27.82M parameters it exceeds the lightweight target. The YOLO26m-cls CE baseline at 89.02% Macro-F1 outperforms all TIMM variants under the same protocol, supporting the selection of YOLO26m-cls as the retained backbone.

0.3.3 Baseline Selection Summary

The YOLO26m-cls CE baseline (89.02% Macro-F1, 89.02% accuracy, 0.8505 Kappa) is the reference point for all subsequent proposed-method comparisons. The screening ceiling of

Table 4: TIMM baseline screening results on SDI-4 test (seed 42).

Model	Params (M)	Macro-F1 (%)	Accuracy (%)
ConvNeXt-Tiny (IN22K)	27.82	84.16	85.22
MobileNetV3-Large	4.21	80.10	82.61
EfficientNet-B0	4.01	77.01	79.13
RepVGG-A0	7.83	76.88	77.39
EfficientNetV2-S	20.18	76.78	77.39
ShuffleNetV2-x1.0	1.26	71.85	74.78
FastViT-T8	3.26	71.03	73.91
EdgeNeXt-XX-Small	1.16	70.63	72.17
MobileOne-S0	4.27	69.69	71.30
MobileViT-S	4.94	69.18	73.91
MobileNetV4-Conv-Small	2.50	66.26	67.83
MNASNet-1.0	3.11	62.48	66.96
GhostNetV2-1.0	4.88	61.01	61.74
ReXNet-1.0	3.52	58.55	61.74
SqueezeNet1.1	0.72	55.04	59.13
MobileNetV4-Hybrid-Medium	9.80	53.83	60.87
EfficientViT-M1	2.79	52.83	57.39

approximately 30M parameters was considered during candidate inclusion, but the final engineering preference favours the 10.35M-parameter YOLO26m-cls for its favourable balance of performance and efficiency.

0.4 Main Classification Performance

This section reports the primary classification results on the SDI-4 test partition. The proposed method (YOLO26m-cls + ASL-LDAM + SimAM-DCFR) is compared against the CE baseline (YOLO26m-cls with cross-entropy loss) under the same training protocol.

0.4.1 Training Configuration

Both the proposed method and the CE baseline were trained on the SDI-4 training partition (804 images) with the following shared configuration:

- Input resolution: 224×224 pixels.
- Epochs: 30, with early stopping patience of 15.
- Batch size: 32.
- Optimiser: AdamW, initial learning rate 0.00125, cosine annealing schedule, final learning rate factor 0.01.

- Mixed precision: AMP enabled.
- Data augmentation: RandAugment, random erasing with probability 0.4.
- Workers: 4.
- Seed: 42.

The proposed method additionally applies:

- **ASL-LDAM**: Asymmetric loss with label-distribution-aware margin. Configuration: $\gamma_{\text{pos}} = 0$, $\gamma_{\text{neg}} = 4$, label smoothing 0.1, LDAM max margin 0.5, LDAM scale 30. Class counts: [282, 139, 229, 154].
- **SimAM-DCFR**: Simple attention module with dual-channel feature recalibration, inserted immediately before the original classification head. The attention branch is parameter-free; the texture branch and channel gate are trainable. Fusion: $\mathbf{F}' = \mathbf{F} + 0.5 \cdot \mathbf{G} \odot (\mathbf{F}_{\text{SimAM}} + \mathbf{F}_{\text{texture}})$, where $\mathbf{G}$ is a learned channel-gating vector. The SimAM energy parameter is $\lambda = 10^{-4}$.

0.4.2 Primary Results

The primary SDI-4 test results are reported in Table 5.

Table 5: Primary SDI-4 classification results (seed 42).

Method	Macro-F1 (%)	Accuracy (%)	Kappa
CE Baseline (YOLO26m-cls)	89.02	89.02	0.8505
Proposed (ASL-LDAM + SimAM-DCFR)	91.01	91.33	0.8816
Δ	+1.99	+2.31	+0.0311

The proposed method improves upon the CE baseline by 1.99 percentage points in Macro-F1, 2.31 percentage points in accuracy, and 0.0311 in Cohen’s Kappa. These improvements are observed under the recorded protocol and seed 42; they support the effectiveness of the ASL-LDAM loss and SimAM-DCFR attention module for this four-class shrimp disease classification task.

0.4.3 Per-Class Analysis

The per-class F1 scores for both methods are available from the retained confusion matrices and classification reports in the experiment archives. The SDI-4 test partition comprises 61 Healthy, 29 BG, 50 WSSV, and 33 WSSV_BG images. The primary sources of confusion are expected between WSSV and WSSV_BG (co-infection boundary) and between Healthy and disease classes (false-positive/negative asymmetry).

0.4.4 Model Capacity and Footprint

The proposed method introduces 529,408 trainable parameters through the SimAM-DCFR module, increasing the total parameter count from 10.35M (YOLO26m-cls backbone) to approximately 10.88M. The FP32 model size of the proposed classifier is approximately 39.52 MiB in TFLite format, as discussed in Section 0.14.

The SDI-4 trained checkpoint hash for the proposed method is: bf4d969465025e870a61be5e218dfc55b9b2e962dec416f220b1f130703fb1c7

The CE baseline checkpoint hash is: 4305e49158129c4a6acaa8fdaf7b5982a7f4d93cc233f11d17479e04b0e438c1

0.5 Classification Component Analysis

The proposed method combines two components: the ASL-LDAM loss function and the SimAM-DCFR attention module. This section examines the contribution of each component through controlled comparisons on the SDI-4 test partition.

0.5.1 Architecture Details

The SimAM-DCFR module is inserted immediately before the original classification head of YOLO26m-cls. It consists of three branches:

- **SimAM (attention) branch:** A parameter-free simple attention mechanism that computes a 3D attention map from the energy of each neuron. The operation is defined as:

$$\mathbf{M}_{\text{SimAM}} = \text{sigmoid}\left(\frac{1}{\hat{\sigma}^2 + \lambda}\right) \odot \mathbf{F},$$

where $\hat{\sigma}^2$ is the variance across spatial positions, $\lambda = 10^{-4}$ is the energy regularisation parameter, and $\mathbf{F}$ is the input feature map.

- **Texture branch:** A trainable branch that captures local texture patterns through learnable filters. The output is $\mathbf{F}_{\text{texture}}$.
- **Channel gate:** A learned gating vector $\mathbf{G}$ that modulates the relative contribution of the attention and texture branches.

The fusion operation is:

$$\mathbf{F}' = \mathbf{F} + 0.5 \cdot \mathbf{G} \odot (\mathbf{F}_{\text{SimAM}} + \mathbf{F}_{\text{texture}}).$$

The SimAM branch is parameter-free, while the texture branch and channel gate introduce trainable parameters. The complete SimAM-DCFR block adds 529,408 trainable parameters to the 10.35M-parameter backbone, resulting in a total of approximately 10.88M parameters. The complete SimAM-DCFR block is therefore not entirely parameter-free; only the SimAM attention computation within the block is parameter-free.

0.5.2 Component Ablation

Table 6 presents the ablation results for the SDI-4 test partition.

Table 6: Component ablation on SDI-4 test (seed 42).

Configuration	Macro-F1 (%)	Accuracy (%)	Kappa
CE baseline (YOLO26m-cls)	89.02	89.02	0.8505
+ ASL-LDAM (no SimAM-DCFR)	—	—	—
+ SimAM-DCFR (CE loss)	—	—	—
+ ASL-LDAM + SimAM-DCFR (full proposed)	91.01	91.33	0.8816

The individual component results (ASL-LDAM-only and SimAM-DCFR-only) involve the ASL-LDAM loss training and attention-benchmark variants. The attention-benchmark experiments were conducted on an RTX 4090, testing nine attention variants under both CE and ASL-LDAM settings. The component isolation results will be populated from the retained experiment logs when the corresponding evidence archive is fully extracted.

0.6 Classification Error Analysis

This section analyses the classification errors made by both the proposed method and the CE baseline on the SDI-4 test partition.

0.6.1 Confusion Pattern

The primary sources of confusion are expected to occur between classes that share visual characteristics. Specific patterns include:

- **WSSV vs. WSSV_BG**: Images exhibiting co-infection may be mistaken for single-infection WSSV when the black gill symptoms are subtle or absent.
- **Healthy vs. disease classes**: False positives on Healthy images (predicting a disease class for a healthy specimen) and false negatives (missing a diseased specimen) carry different operational implications.
- **BG vs. WSSV_BG**: The presence of black gill symptoms in both classes creates a natural boundary ambiguity.

Detailed confusion matrices are available from the retained experiment logs and will be reported in the final thesis.

0.6.2 Error Reduction

The overall accuracy improvement from 89.02% (CE baseline) to 91.33% (proposed) corresponds to approximately 4 fewer misclassified images out of the 173-image test set. The component-level contribution to error reduction will be analysed when the individual component logs are extracted.

0.7 Classification Robustness

This section examines the robustness of the proposed classifier under controlled variations. The analysis is based on the recorded test protocol and does not claim universal robustness beyond the evaluated conditions.

0.7.1 Data Augmentation Sensitivity

Both the proposed method and the CE baseline were trained with RandAugment and random erasing (probability 0.4). The impact of individual augmentation strategies was not isolated in the current protocol; the observed performance reflects the combined augmentation pipeline.

0.7.2 Inference Robustness

The proposed classifier was evaluated under the SDI-4 test partition without additional inference-time augmentation. The reported results (Macro-F1 91.01%, accuracy 91.33%) therefore represent the expected performance under standard evaluation conditions. Extended robustness testing, including test-time augmentation (TTA) and input perturbation, is a direction for future work.

0.8 Additional-Source and Integrated Evaluation

This section evaluates the proposed classifier on the EXT-3-Original dataset (three classes, external source) and the integrated SDI-4 + EXT-3-Original benchmark. The integrated evaluation exposes the classifier to domain shift and class-composition changes, providing a more challenging generalisation test.

0.8.1 EXT-3-Original Performance

The EXT-3-Original dataset contains 315 images across three classes (Healthy, BG, WSSV) with a fixed partition of 221/47/47. Results are reported in Table 7.

The proposed method achieves a substantial improvement over the CE baseline on the EXT-3-Original dataset, with a Macro-F1 of 91.29% compared to 72.30%. This improvement is larger than the improvement observed on SDI-4, suggesting that the ASL-LDAM +

Table 7: EXT-3-Original classification results (seed 42).

Method	Accuracy (%)	Macro-F1 (%)
CE Baseline	70.21	72.30
Proposed	91.49	91.29
Δ	+21.28	+18.99

SimAM-DCFR components are particularly beneficial when the class distribution shifts or the acquisition conditions differ from the primary training source.

0.8.2 Integrated SDI-4 + EXT-3-Original Performance

The integrated dataset combines SDI-4 and EXT-3-Original at the partition level, yielding 1,025 training, 219 validation, and 220 test images. The test partition contains images from both sources, with the WSSV_BG class appearing only in the SDI-4 subset. Results are reported in Table 8.

Table 8: Integrated SDI-4 + EXT-3-Original classification results (seed 42).

Method	Accuracy (%)	Macro-F1 (%)
CE Baseline	87.73	86.51
Proposed	83.18	82.18

0.8.3 Discussion of the Negative Integrated Result

The integrated evaluation reveals a notable negative result: the proposed method (82.18% Macro-F1) underperforms the CE baseline (86.51% Macro-F1) on the combined dataset. This result is a genuine observation under the recorded protocol and must be clearly acknowledged.

The source-stratified integrated accuracy provides additional insight:

- Proposed EXT subset accuracy: 85.11%
- Proposed SDI subset accuracy: 82.66%
- CE EXT subset accuracy: 91.49%
- CE SDI subset accuracy: 86.71%

Both methods degrade on the SDI subset relative to the SDI-4-only evaluation, indicating that the domain shift introduced by EXT-3-Original affects the four-class decision boundary. The degradation is more pronounced for the proposed method, which achieved 91.01% Macro-F1 on SDI-4 alone but only 82.66% on the SDI subset of the integrated test.

Several hypotheses may explain this degradation, but they remain hypotheses rather than established causal explanations:

- **Source interaction:** The combined training set introduces images from different acquisition conditions, potentially shifting the feature distribution learned by the backbone.
- **WSSV vs. WSSV_BG boundary:** The integrated evaluation includes only SDI-4 images for WSSV_BG, while the WSSV class receives additional samples from EXT-3-Original. This asymmetry may alter the LDAM margin computation.
- **Loss dynamics:** The ASL-LDAM loss, which is sensitive to class priors, may behave differently when the prior distribution changes through the addition of EXT-3 samples.
- **Heterogeneous texture:** The SimAM-DCFR texture branch may overfit to source-specific textures when the training distribution becomes more heterogeneous.

This negative result does not invalidate the primary SDI-4 findings but establishes a clear limitation: the proposed method’s multi-source generalisation capability is condition-dependent and requires further investigation. The CE baseline, while weaker on the individual SDI-4 and EXT-3 benchmarks, shows greater robustness to the integrated training condition.

0.9 Explainability Analysis

Explainability analysis aims to provide visual evidence that the classifier attends to disease-relevant regions rather than background artefacts. The retained evidence includes attention maps and feature visualisations from the SimAM-DCFR module.

0.9.1 Attention Visualisation

The parameter-free SimAM branch produces a 3D attention map that highlights salient spatial locations. The attention weights are computed from the energy distribution of each neuron, with higher energy indicating greater informational relevance. Preliminary visualisations suggest that the attention maps activate around shrimp body regions for disease-positive samples, but a systematic quantitative XAI evaluation (e.g. Grad-CAM, occlusion sensitivity) remains as future work.

0.9.2 Limitations

The current XAI analysis is qualitative. The following limitations apply:

- No ground-truth attribution masks are available for the shrimp disease domain.

- Attention maps from the SimAM module do not necessarily correspond to causal decision regions.
- A systematic comparison of XAI methods (e.g. Grad-CAM, Integrated Gradients, LIME) has not been conducted.

The explainability discussion is therefore confined to observations from the available attention maps and does not claim causal interpretation.

0.10 Segmentation Baselines and Leakage Analysis

Segmentation experiments were conducted by team member Nguyen Van Phong. This section reproduces the baseline segmentation results from the accepted evidence. The classification agent does not modify the underlying segmentation claims.

0.10.1 Baseline YOLO11n-Seg

The baseline segmentation model uses YOLO11n-seg trained on the shrimp disease segmentation dataset. The dataset includes polygon annotations for two positive classes: BG (black gill) and WSSV (white spot). Healthy images from the original ShrimpDiseaseImageBD source are retained in the segmentation training set as negative samples (empty-label images). These images contribute zero-valued mask loss during optimisation but do not introduce positive mask supervision for the Healthy class.

0.10.2 Leakage Consideration

An important semantic distinction is that Healthy is not a positive segmentation mask class. However, residual false-positive disease masks on Healthy images motivate the use of classification-gated segmentation as an additional operational safeguard. The cascade risk is that a diseased image misclassified as Healthy by the classifier may suppress segmentation, so end-to-end cascade sensitivity remains a required validation boundary for any integrated deployment.

0.10.3 Baseline Results

Detailed baseline segmentation metrics (mAP@50, mAP@50:95, and per-class results) are reported in the retained segmentation experiment logs. The classification agent reproduces these results from the accepted baseline without modification.

0.11 Attention-Based Segmentation

Attention-based segmentation experiments build on the YOLO11n-seg baseline by incorporating attention mechanisms into the segmentation backbone. These experiments were conducted by Nguyen Van Phong, and the results are reproduced from the accepted evidence.

0.11.1 Attention Variants

The evaluated attention variants include modifications to the segmentation head and backbone to incorporate channel-wise and spatial attention. The specific attention configurations and their corresponding segmentation metrics are documented in the retained segmentation experiment logs.

0.11.2 Results

The attention-based segmentation results are reported in the retained evidence. The classification agent reproduces these results from the accepted baseline and does not introduce new claims.

0.12 Fourier-Domain Segmentation

Fourier-domain segmentation experiments investigate the use of frequency-domain representations for shrimp disease segmentation. These experiments were conducted by Nguyen Van Phong, and the results are reproduced from the accepted evidence.

0.12.1 Motivation

Fourier-domain features can capture periodic texture patterns that are characteristic of certain disease manifestations. The Fourier spectrum analysis was applied to the segmentation feature maps before the mask head, and the impact on segmentation accuracy was measured.

0.12.2 Results

The Fourier-domain segmentation results are documented in the retained experiment logs. The classification agent reproduces these results from the accepted evidence and does not introduce new claims.

0.13 Segmentation Robustness and Failure Analysis

Segmentation robustness analysis documents known failure modes and limitations of the segmentation models. These experiments were conducted by Nguyen Van Phong, and the findings are reproduced from the accepted evidence.

0.13.1 Known Failure Modes

The following failure modes have been documented in the retained segmentation evidence:

- False-positive disease masks on Healthy images, where the segmentation model produces spurious BG or WSSV polygons despite the absence of disease.

- Under-segmentation of WSSV lesions on specimens with mild or early-stage infection.
- Boundary ambiguity between BG and WSSV regions in co-infection cases.

0.13.2 Operational Implications

The false-positive failure mode on Healthy images motivates the classification-gated cascade architecture described in Section 0.10.2. However, the cascade also introduces a failure mode of its own: a diseased image that is misclassified as Healthy by the classifier will suppress segmentation entirely, potentially missing a detectable disease. End-to-end validation of the cascade is therefore necessary before any operational deployment.

0.14 Mobile Deployment

Mobile deployment experiments provide preliminary evidence of the proposed classifier’s feasibility on mobile devices. The results are treated as prototype evidence; they do not constitute clinical or real-time deployment guarantees.

0.14.1 TFLite Conversion

The trained YOLO26m-cls + ASL-LDAM + SimAM-DCFR model was converted to TFLite format in two variants:

- **FP32:** Approximately 39.52 MiB. File: yolo26m_asl_ldam_simam_dcf_r_fp32.tflite.
- **FP16:** Approximately 19.81 MiB. File: yolo26m_asl_ldam_simam_dcf_r_fp16.tflite.

The FP16 variant uses reduced precision for storage and computation. It is not referred to as ‘quantized’ in the protocol sense of integer quantization. The FP32 variant is full-precision floating point and is not quantized.

0.14.2 GPU Inference (Kaggle T4)

GPU-accelerated inference benchmarks were conducted on a Kaggle T4 GPU. The retained evidence includes latency measurements and throughput results. These measurements are recorded under controlled conditions and do not reflect mobile-device inference performance.

0.14.3 Limitations

The following limitations apply to the mobile deployment evidence:

- No on-device mobile inference measurements (e.g. Android, iOS) are currently available.

- T4 GPU FPS cannot be used as mobile FPS.
- The TFLite delegates have not been validated on target mobile hardware.
- Real-time performance has not been established.
- The prototype is not clinically ready.

Mobile deployment remains a direction for future work, as discussed in Section 0.17.

0.15 Integrated Discussion

This section synthesises the classification and segmentation findings presented in this chapter.

0.15.1 Classification Summary

The proposed ASL-LDAM + SimAM-DCFR method achieves a Macro-F1 of 91.01% on the SDI-4 test partition, representing an observed improvement of 1.99 percentage points over the CE baseline. The improvement is consistent across the primary four-class benchmark. However, the integrated evaluation reveals a condition-dependent degradation (82.18% vs. 86.51% Macro-F1), indicating that multi-source generalisation capability is not guaranteed.

0.15.2 Segmentation Summary

Segmentation experiments, conducted by Nguyen Van Phong, demonstrate the feasibility of BG and WSSV mask prediction. The Healthy class is not a positive segmentation label, but Healthy images are retained as negative training samples. Residual false-positive disease masks on Healthy images motivate classification-gated segmentation as an operational safeguard.

0.15.3 Toward an Integrated System

An integrated shrimp disease detection system would combine the proposed classifier with a segmentation model in a cascade: the classifier provides a coarse disease decision, and the segmentation model refines the localisation when a disease is detected. The cascade introduces a failure mode—a diseased image misclassified as Healthy may suppress segmentation—so end-to-end validation remains a required boundary for any deployment.

0.16 Threats to Validity

This section identifies threats to the validity of the experimental findings and the boundaries within which the claims are supported.

0.16.1 Internal Validity

- **Single-seed evaluation:** All experiments were conducted under a single fixed seed (42). The observed improvements may not generalise across random seeds.
- **Hyperparameter dependence:** The ASL-LDAM and SimAM-DCFR hyperparameters were tuned on the SDI-4 validation set. The reported performance may be optimistic relative to an unbiased estimate.
- **Component interaction:** The individual contributions of ASL-LDAM and SimAM-DCFR have not been fully isolated in the current analysis. The component ablation table will be populated when the corresponding evidence is fully extracted.

0.16.2 External Validity

- **Dataset scope:** The SDI-4 dataset contains 1,149 images from a single primary source. The EXT-3-Original dataset adds 315 images from an external source. The combined dataset size is modest by deep learning standards.
- **Acquisition conditions:** The images were captured under controlled laboratory or known acquisition conditions. Generalisation to uncontrolled field conditions has not been established.
- **Mobile deployment:** Mobile inference evidence is preliminary and does not establish real-time or clinical deployment readiness.

0.16.3 Construct Validity

- **Metric choice:** Macro-F1 is the primary metric, treating all classes equally. In an operational setting, different misclassification costs may apply (e.g. missing WSSV is more costly than false-alarming Healthy).
- **Bounding claims:** Claims are bounded by the recorded protocol and the available evidence. The study does not establish universal robustness, causal attribution, or clinical diagnostic capability.

0.17 Recommendations

Based on the findings presented in this chapter, the following recommendations are made for future work:

0.17.1 Methodological Extensions

- **Multi-seed evaluation:** The current single-seed results should be confirmed with additional random seeds to establish statistical significance.
- **Component ablation:** The individual contributions of ASL-LDAM and SimAM-DCFR should be isolated and reported with full attention-benchmark logs.
- **Multi-source training strategies:** The observed degradation in the integrated condition suggests that domain-aware training strategies (e.g. domain adversarial training, mixup across sources, or source-weighted loss) should be investigated.

0.17.2 Coverage Expansion

- **Field data collection:** The current dataset does not capture the full diversity of real-world shrimp farming conditions. Collaboration with aquaculture facilities to collect field data is a priority.
- **Mobile deployment validation:** On-device inference on representative mobile hardware (Android, iOS) is necessary to validate the TFLite deployment path.
- **Real-time testing:** End-to-end latency and throughput measurements on target hardware should be conducted under realistic conditions.

0.17.3 Validation

- **End-to-end cascade validation:** The classification-gated segmentation cascade should be validated as a complete system, with particular attention to the failure mode where a diseased image is misclassified as Healthy.
- **XAI verification:** Systematic XAI evaluation (Grad-CAM, occlusion) should be conducted to verify that the classifier attends to disease-relevant regions.

0.18 Personal Reflection

This project has been a significant learning experience in applying deep learning to a real-world agricultural challenge. The work involved extensive experimentation with classification architectures, loss functions, and attention mechanisms, as well as collaboration on the segmentation component.

The primary challenge was balancing model capacity with the lightweight target for mobile deployment. The YOLO26m-cls backbone at 10.35M parameters proved to be an effective choice, and the ASL-LDAM + SimAM-DCFR components contributed measurable improvements. However, the negative integrated result and the preliminary nature of the mobile evidence serve as humbling reminders that field deployment remains a distant goal.

The project also highlighted the importance of disciplined experimental practice: machine-readable result logging, checkpoint hashing, and clear provenance tracking were essential for producing a reproducible evaluation. Future work should build on this foundation with broader validation, multi-seed evaluation, and real-world deployment testing.